

The elusive nature of early phonological priming effects: Are there individual differences?[☆]

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Abstract

Lukatela and Turvey (2000) demonstrated a phonological priming effect in the lexical decision task (LDT) with a 14-ms prime and concluded that phonology plays a central role in word meaning activation. In contrast, several other researchers reported that phonological priming is significant only at much longer prime durations (e.g., Ferrand & Grainger, 1994). In two replication attempts (Experiments 1a and 1b), involving a 15-ms prime duration, we found a clear phonological priming effect in one LDT and no evidence of phonological priming in another virtually identical LDT. In Experiment 2, in an attempt to determine whether individual differences may account for the presence or lack of a phonological priming effect, we also tested phonological and perceptual skill. Only participants higher in perceptual and phonological skill showed a phonological priming effect. We conclude that these (and potentially other) variables may have been responsible for previous inconsistent findings of early phonological priming effects. © 2003 Elsevier Inc. All rights reserved.

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1. Introduction

Most theories of word recognition assume that some form of phonological representation is generated when reading text silently. There is, however, no agreement as to whether the phonology that is generated in silent reading is assembled phonology or addressed phonology. Assembled phonology is a phonological representation of a word that is computed quickly and leads to the activation of word meaning. Addressed phonology is the phonological representation of a word that is generated after the meaning of that word has already been accessed. The main point of contention among theorists is whether a phonological representation is generated prior to word meaning activation. The issue of when, in

the course of visual word recognition, a phonological representation is generated is also the focus of the present research.

The idea that the sound of language is generated when we read makes intuitive sense. Infants first encounter language verbally by listening to the speech of others. Only later does acquiring the skill of reading give the child the ability to represent the sounds of language using an arbitrary orthographic (written) system. Phonological awareness in children has been shown to be one of the best predictors of later reading achievement and can be used to differentiate between good and poor readers (Torgesen, Wagner, Rashotte, Burgess, & Hecht, 1997). As well, dyslexics, who have a pronounced reading deficit, also tend to have a deficit in their phonological skills (e.g., Bruck, 1990; Bruck, 1992; Fawcett & Nicolson, 1995).

1.1. Theories of visual word recognition

The discussion of the role phonology plays in visual word recognition has involved two opposing theoretical camps. These theories differ in the relative importance

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they place on the role of phonological versus orthographic processing during word recognition.

One class of models postulates two different routes for word recognition. According to this dual-route framework, one route uses orthographic information to directly access the meaning of a word. The other route is a phonologically mediated pathway in which the assembly of phonology is an intermediate step between orthography and word meaning activation. The dual-route models assume that the direct access orthographic route is the faster of the two routes because it bypasses the computation of phonology. Word recognition for high frequency words has already occurred by the time the phonological recoding process is engaged, allowing for only minimal influence of the phonological route on the word recognition process. The role of phonology is thought to be limited to the recognition of nonwords and low-frequency words that require the computation of phonology because their orthographic representation has been encountered rarely, or never before (Coltheart, 1978; Coltheart, Besner, Jonansson, & Davelaar, 1979; Coltheart, Curtis, Atkins, & Haller, 1993; McCusker, Hillinger, & Bias, 1981; Paap & Noel, 1991; Plaut, McClelland, Seidenberg, & Patterson, 1996; Seidenberg, 1985; Seidenberg & McClelland, 1989; Seidenberg, Waters, Barnes, & Tanenhaus, 1984).

An alternate theory of visual word recognition states that phonology plays a central role in the recognition of words. Frost (1998; see also Van Orden et al., 1990) has proposed a strong phonological theory, according to which phonological processing is the primary mode by which word recognition occurs. Whereas the dual-route models assume that the computation of phonology is a procedure that is relevant only when a particular letter string has never been encountered before, the strong phonological model posits that the computation of phonology is the default procedure of the word recognition system.

1.2. Evidence from phonological priming

One of the testable predictions of the strong phonological model is that phonological processing is engaged rapidly, in the earliest stages of visual word recognition. This prediction is opposite to the dual-route view that the computation of phonology is slow and unnecessary, except in rare circumstances.

Several research paradigms have been designed to test whether the computation of phonology occurs early in the word recognition process. Many of the studies that have found evidence for the computation of phonology (Lukatela & Turvey, 1990, 1991, 1993, 1994a, 1994b) have used experimental tasks that require the production of phonology (e.g., overt naming tasks). Naming tasks have been criticized, however, because they involve pronunciation of the

word and thus absolutely require the computation of phonology.

Other tasks, like the perceptual identification task and the lexical decision task (LDT), avoid this problem. With these tasks, the masked priming paradigm has been used to investigate early processing. Masked priming is a technique wherein a briefly presented prime and target are both preceded by pattern masks (e.g., XXXX) that disrupt their processing. Studies involving perceptual identification tasks (Berent & Perfetti, 1995; Grainger & Ferrand, 1994, 1996; Perfetti & Bell, 1991; Perfetti, Bell, & Delaney, 1988), in which the participant attempts to identify a briefly presented target by indicating what he/she saw, and studies involving LDTs (Berent, 1997; Ferrand & Grainger, 1993; Grainger & Ferrand, 1994; Lukatela & Turvey, 2000; Lukatela, Frost, & Turvey, 1998) have produced results demonstrating facilitation (faster RTs) when a prime shares greater phonological overlap with a target (e.g., *KLIP-clip*) in comparison to a prime that shares the same amount of orthographic overlap but less phonological overlap (e.g., *CLEP-clip*). Evidence for early phonological activation using masked priming has been demonstrated in English and also in Hebrew (Frost & Yogeve, 2001) and Dutch (Brysbaert, 2001).

The results of several studies by Ferrand and colleagues (Ferrand & Grainger, 1992, 1993, 1994; Grainger & Ferrand, 1996) suggest that orthographic information provides the earliest constraint on word processing, while phonological information is only computed at a later time (i.e., prime duration of 50 ms or longer). The same pattern of results was replicated by Ziegler, Ferrand, Jacobs, Rey, and Grainger (2000) in an incremental priming task.

Other research using the backward masking paradigm in perceptual identification tasks has demonstrated evidence for phonological assembly within the first 40 ms of word recognition (Perfetti & Bell, 1991; Perfetti et al., 1988). In the backward masking procedure, participants view a briefly presented target word (e.g., *clown*) that is immediately followed by a letter string mask (e.g., *CLAUNE*). In these experiments, nonwords that were phonologically (e.g., *HEDD*; referred to as pseudohomophones) similar to the targets (e.g., *head*) they masked resulted in faster and more accurate word identification than did orthographically similar controls (e.g., *HEED*). Most importantly, these studies showed evidence for the computation of phonology in tasks that did not require explicit pronunciation.

1.3. The present research

Recently, Lukatela and colleagues have produced results demonstrating not only significant phonological priming in lexical decision, but phonological priming

independent of orthographic similarity (e.g., responses to KLIP–*clip* are faster than to CLOP–*clip*) using 57, 29, and even 14-ms primes (Lukatela et al., 1998; Lukatela & Turvey, 2000). Based on these results, Lukatela and colleagues have argued for a model of word recognition in which phonological assembly plays the leading role.

In light of these claims, which have important implications for models of word recognition, other researchers attempted to replicate the results of Lukatela et al. (1998). Independently, Forster and Guerrerá-Mahoney (submitted) and Coltheart and Woollams (personal communication, January 22, 2003) found no pseudohomophone priming effect with a 55-ms prime duration. That is, RTs were equivalent for targets (*clip*) presented with pseudohomophone primes (KLIP) and targets presented with nonhomophonic primes (CLOP) where orthographic overlap was the same for both types of prime–target pairings. These two independent failures to replicate suggest that evidence for early activation of phonology is not robust and perhaps that there are some experimental conditions that facilitate phonological priming and some that do not. Other researchers were also unable to find significant phonological priming with similar prime durations (Bowers, Vigliocco, & Haan, 1998; Davis, Castles, & Iakovidis, 1998).

Although two independent failures to replicate Lukatela et al. (1998) cast some doubts on the generalizability of the results, Lukatela and Turvey (2000) not only replicated their own finding of early phonological priming but found it with an extremely short prime duration of 14 ms. Lukatela and Turvey used a masked priming procedure in a LDT. Across several experiments, these authors found a significant priming advantage at a 14-ms prime duration for targets presented with pseudohomophone primes (e.g., KREST–*crest*) compared to targets presented with nonword primes that conveyed the same amount of orthographic information (e.g., CROST–*crest*). It is important to note that in this study Lukatela and Turvey only found evidence of early phonological assembly for words with a simple rather than a complex vowel structure. They concluded that vowel complexity might play a role in the early activation of phonological information.

The purpose of the present Experiments 1a and 1b was to attempt to replicate Lukatela and Turvey's (2000) significant pseudohomophone priming effect with a 14-ms prime duration. We attempted to create as much similarity as possible between the present experiments and those of Lukatela and Turvey (2000) in order to minimize the possibility that any differences in results could be attributed to differences in methodology. It is important to attempt to replicate such results because the claim that phonological assembly plays the leading role in word recognition is not widely accepted and has important implications for models of word recognition.

2. Experiment 1

The purpose of Experiments 1a and 1b was to examine whether pseudohomophone primes provide significantly greater priming in comparison to primes that share the same amount of orthographic overlap but less phonological overlap with targets. One aspect of our procedure that differed from Lukatela and Turvey's (2000) was the inclusion of an unrelated prime condition. Lukatela and Turvey did not include an unrelated condition, but without this condition it is difficult to know whether primes are having any effect on target word processing, unless the advantage for pseudohomophone primes is observed. Orthographic priming would be indicated by significantly faster response times to targets with orthographically similar primes compared to targets with unrelated primes that share no orthographic or phonological overlap.¹ Phonological priming would be indicated by significantly faster response times to targets preceded by pseudohomophone primes (e.g., KREST–*crest*) as compared to targets preceded by primes that share the same amount of orthographic similarity as the pseudohomophone primes but less phonological overlap (e.g., CROST–*crest*).

Lukatela and Turvey (2000) found differential priming for pseudohomophone and orthographic control primes depending on the phonological and orthographic complexity of the target words (monophthongs and diphthongs). Significant phonological priming above and beyond orthographic priming was observed only for monophthongs (simple vowel) and not for diphthongs (complex vowel). They hypothesized that it was the added phonological complexity of the vowel in diphthong targets that slowed down phonological processing to the extent that there was no difference between the pseudohomophone and orthographic control prime conditions.

Lukatela and Turvey's (2000) word list was constructed in such a way that it is difficult to evaluate orthographic priming effects for monophthong vs. diphthong targets. That is, there is a difference in the similarity of the orthographic control primes and targets in the monophthong and the diphthong stimulus sets. The orthographic primes for the monophthong targets are different from the targets by only one letter (e.g.,

¹ We are assuming that any benefit produced by primes like CROST (for the target *crest*) relative to unrelated primes can be attributed primarily to orthographic processing. It is certainly the case, however, that these orthographic primes have some degree of phonological overlap with the targets. That is, the prime and target in this example share consonants and also share consonant phonology. Although effects of this type of prime are not *purely* orthographic effects, we will call them orthographic since this choice of terminology facilitates comparison to Lukatela and Turvey's studies, and because these primes do have less phonological overlap with targets than do the pseudohomophone primes.

CROST–*crest*). In contrast, the orthographic primes for the diphthong targets are different from the targets by two letters (e.g., JONK–*joke*). Therefore, taking into account letter position, there is a greater amount of orthographic overlap between primes and targets for the monophthong stimuli than for the diphthong stimuli. This reduces the likelihood of finding orthographic priming for the diphthong targets. In order to be able to evaluate whether there was comparable orthographic priming for monophthong and diphthong targets, in the present Experiment 1b new orthographic primes were created for the diphthong target words that differed by only one letter (e.g., JOKN–*joke*) as opposed to two letters (e.g., JONK–*joke*).² It was not possible to make the same change to the pseudohomophone primes for the diphthong targets and still preserve their pseudohomophony. Other than this change to the orthographic control primes for the diphthong stimuli, Experiments 1a and 1b were identical.

2.1. Method

2.1.1. Participants

Thirty University of Calgary undergraduates participated in Experiment 1a and 39 participated in Experiment 1b. These students received bonus credit toward a psychology course in exchange for participation. All participants reported that English was their first language. Different participants took part in each of the separate experiments.

2.1.2. Materials

The stimulus list used was obtained from Appendices A and B of Lukatela and Turvey (2000) and was used in their Experiment 1 (see present Appendix A). There were 192 trials in total: 96 word trials and 96 nonword trials. The 96 target words were divided in half such that 48 words were monophthongs (simple vowels) and 48 were diphthongs (complex vowels). Each target word was a monosyllable that was four to five letters in length. The mean frequency for monophthong target words was 23.96 ± 38.85 and the neighborhood density was 8.69 for target words and 18.58 for word bodies. The mean frequency for diphthong target words was 95.48 ± 148.35 and the mean neighborhood density was 13.21 for target words and 13.31 for word bodies. An unrelated prime that shared no phonological or orthographic overlap

was created for each target word. Thus, each target word (e.g., *crest*) had an associated pseudohomophone prime (e.g., KREST), an orthographic control prime (e.g., CROST), and an unrelated prime (e.g., GOAMP). The word list was divided into three separate lists (Lists A, B, and C) so that no participant saw the same target word twice.

2.1.3. Procedure

The experimental paradigm involved a modified LDT. Based on arguments made by Berent and Perfetti (1995) about early phonological assembly and LDT, Lukatela and Turvey (2000) modified the LDT to only present the target word for a specified, brief period of time. Participants were presented with strings of white letters on a black computer screen. The experimental paradigm requires the use of extremely rapid presentation rates for the mask, prime, and target. Therefore, the amount of luminance from the computer screen, and in the testing room in general, becomes extremely important. As in Lukatela and Turvey's study, the lights in the testing room were dimmed so that the ambient luminance of the room was low (0.9 cd/m^2 in the present experiments). The luminance from the computer screen was 22.4 cd/m^2 . Participants sat at a viewing distance of approximately 55 cm. Lastly, the refresh cycle of the computer screen was 15 ms.

In each trial, there were three events. The first event was a pattern mask presented for 500 ms. The pattern mask (____) was a line that appeared in the character spaces in which the letter string subsequently appeared. Lukatela and Turvey (2000) argued that this type of mask, which differs from the usual pattern mask (e.g., &&&&&), was appropriate because it functioned as a metacontrast mask. They argued that the advantage of the nonoverlapping forward mask is that it minimizes the integrative phase of the mask and prime that occurs at extremely short presentation rates.

The second event was the prime (e.g., KREST). The prime was a letter string presented for 15 ms. Lastly, the target (e.g., *crest*) was presented for 130 ms. Note that Lukatela and Turvey (2000) used a target presentation duration of 72 ms, but in pilot testing we found that, for our participants, target classification (i.e., lexical decision) was at chance for this presentation duration. The 130 ms target duration was chosen after pilot testing which revealed that at 130 ms the error rate was still somewhat high but considerably less than chance. A blank screen followed the target and remained until the participant responded. Immediately after the response, there was a 1500 ms interstimulus interval and then the next trial. Reaction time was recorded as the time from the presentation of the target word to the time of the participant's response. As in Lukatela and Turvey's experiments, the prime was presented in uppercase and the target was presented in lowercase.

² This change to the stimuli enhances the orthographic overlap between orthographic control primes and diphthong targets. It also tends to make the orthographic control primes less pronounceable. Consequently, the revised orthographic control primes for diphthong targets used in Experiments 1b and 2 should more clearly have an orthographic (and not phonological) effect on processing. When the effects of these orthographic control primes are compared to the effects of the pseudohomophone primes, the contribution of phonological processing should be better isolated.

2.2. Results

2.2.1. Experiment 1a

Mean reaction times and mean error percentages for Experiment 1a are presented in Table 1. The mean error percentage for responses to word targets was 9%. Planned comparisons were conducted between prime conditions for mean reaction times and error percentages for both the monophthong and diphthong stimuli. Results of these planned comparisons are also presented in Table 1.

2.2.2. Experiment 1b

Mean reaction times and mean error percentages for Experiment 1b are presented in Table 2. The mean error percentage for responses to word targets was 14%. Planned comparisons were conducted between prime conditions for mean reaction times and error percentages for both the monophthong and diphthong stimuli. Results of these planned comparisons are also presented in Table 2.

2.3. Discussion

In Experiment 1a, for both monophthong and diphthong target words there was significant priming for both types of related primes in comparison to the unrelated prime. There was, however, no advantage of pseudohomophone primes over orthographic primes so the observed priming is likely orthographic.

In contrast, the results of Experiment 1b do replicate the pattern of findings reported by Lukatela and Turvey

(2000) with respect to the differential priming by pseudohomophones and orthographic controls for words with simple and complex vowels. That is, for monophthong targets, both the pseudohomophone and the orthographic primes provided significantly more priming than the unrelated control prime. However, unlike Experiment 1a, in which a pseudohomophone priming effect for monophthong targets was not replicated, the pseudohomophone primes in Experiment 1b provided significant facilitation above and beyond the orthographic primes. This is the opposite result to that observed in Experiment 1a, even though exactly the same primes and monophthong target stimuli were used. The findings of the present Experiments 1a and 1b are contrasting results that lead to opposite theoretical conclusions regarding the role of phonology in early word processing. This serves to highlight apparent volatility in the masked priming paradigm with such brief presentation durations for the prime and target. All of the conditions in these two experiments were the same (except for a slight change to the orthographic primes used with diphthong targets) and yet completely different results were obtained.

For diphthong targets in Experiment 1b, there was significant facilitation for the pseudohomophone prime condition over the unrelated prime condition, and for the orthographic prime condition over the unrelated prime condition. Given the change to the orthographic control primes for diphthong targets, however, it is difficult to interpret the difference between the pseudohomophone prime and orthographic control prime conditions. The pseudohomophone primes in this case

Table 1
Mean reaction times (in ms), error percentages, and priming effects for Experiment 1a as a function of prime type

Prime type	Target vowel complexity	RT		Error	
		Mean	SE	Mean	SE
Pseudohomophone	M	562	18.29	7.96	1.13
	D	566	15.97	7.56	1.06
Orthographic control	M	547	13.21	6.89	1.02
	D	572	14.98	7.80	1.22
Unrelated	M	591	15.86	13.58	2.09
	D	595	20.18	11.51	2.01
<i>Priming effects</i>					
Pseudohomophone vs. Orthographic	M	+15	$t_1(29) = 2.02, p < .05$ $t_2(47) = 1.25, p > .05$	+1.07	Both $t < 1$
	D	−6	Both $t < 1$	−0.24	$t_1(29) = -1.24, p > .05$ $t_2 < 1$
Orthographic vs. Unrelated	M	−44	$t_1(29) = -6.88, p < .001$ $t_2(47) = -4.10, p < .001$	−6.69	$t_1(29) = -3.00, p < .01$ $t_2(47) = -3.51, p < .001$
	D	−23	$t_1(29) = -1.82, p < .05$ $t_2(47) = -1.49, p > .05$	−3.71	$t_1(29) = -2.77, p < .01$ $t_2(47) = -2.11, p < .05$
Pseudohomophone vs. Unrelated	M	−29	$t_1(29) = -2.94, p < .01$ $t_2(47) = -2.57, p < .05$	−5.62	$t_1(29) = -4.57, p < .001$ $t_2(47) = -3.20, p < .01$
	D	−29	$t_1(29) = -2.67, p < .01$ $t_2(47) = -2.24, p < .05$	−3.95	$t_1(29) = -3.41, p < .01$ $t_2(47) = -2.80, p < .01$

Note. RT, reaction time; SE, standard error; M, monophthong; and D, diphthong.

Table 2

Mean reaction times (in ms), error percentages, and priming effects for Experiment 1b as a function of prime type

Prime type	Target vowel complexity	RT		Error	
		Mean	SE	Mean	SE
Pseudohomophone	M	558	11.75	10.26	1.21
	D	578	11.86	11.91	1.40
Orthographic control	M	585	17.54	10.87	1.99
	D	561	14.12	9.64	1.96
Unrelated	M	633	15.25	18.28	2.41
	D	601	15.90	20.40	3.10
<i>Priming effects</i>					
Pseudohomophone vs. Orthographic	M	–27	$t_1(38) = -2.35, p < .05$ $t_2(47) = -1.80, p = .07$	–0.61	Both $t < 1$
	D	+17	$t_1(38) = 1.35, p > .05$ $t_2(47) = 1.12, p > .05$	+2.27	$t_1(38) = 1.20, p > .05$ $t_2 < 1$
Orthographic vs. Unrelated	M	–48	$t_1(38) = -3.59, p < .01$ $t_2(47) = -2.83, p < .01$	–7.41	$t_1(38) = -4.33, p < .001$ $t_2(47) = -3.19, p < .01$
	D	–40	$t_1(38) = -4.12, p < .001$ $t_2(47) = -3.32, p < .01$	–10.76	$t_1(38) = -4.23, p < .001$ $t_2(47) = -4.45, p < .001$
Pseudohomophone vs. Unrelated	M	–75	$t_1(38) = -5.57, p < .001$ $t_2(47) = -4.54, p < .001$	–8.02	$t_1(38) = -5.06, p < .001$ $t_2(47) = -3.41, p < .01$
	D	–23	$t_1(38) = -2.50, p < .01$ $t_2(47) = -2.22, p < .05$	–8.49	$t_1(38) = -3.21, p < .01$ $t_2(47) = -3.09, p < .01$

Note. RT, reaction time; SE, standard error; M, monophthong; and D, diphthong.

were less orthographically similar to the targets than were the orthographic control primes. This resulted in a 17-ms disadvantage for responses for targets in the pseudohomophone prime condition compared to the orthographic prime condition. This orthographic disadvantage could, in theory, have been offset by a phonological priming effect, but the observed tendency toward a disadvantage makes it seem unlikely that phonological priming was affecting responses for diphthong targets. Our conclusion for the diphthong targets is that there was evidence for orthographic processing but not phonological processing.

3. Experiment 2

The results of Experiment 1 demonstrate that within one laboratory, using virtually identical materials, it is possible to both fail to replicate early phonological priming effects for monophthong targets and to successfully replicate early phonological priming effects for monophthong targets. The results of Experiment 1a are consistent with those of others who have been unable to replicate an early pseudohomophone priming effect and suggest that evidence for phonological assembly can only be found with much longer prime durations. In contrast, the results of Experiment 1b replicate those of Lukatela and Turvey (2000) and suggest that phonological assembly occurs quickly, at least for monophthongs, and plays a leading role in visual word recognition. Clearly, both of these conclusions cannot be entirely correct. Therefore, it seems possible that

there exists some other variable that mediates whether or not participants will demonstrate a phonological priming effect. We next consider what that variable might be.

It has been suggested that differences in phonological priming may be due to conscious awareness of primes. Koudier (2002) claimed that phonological priming was restricted to participants who were consciously aware of the primes. This explanation is, however, inconsistent with reports of phonological priming with 14-ms prime exposures (Lukatela & Turvey, 2000, the present Experiment 1b) since it is highly unlikely that participants would be able to identify primes presented for such brief durations.

Davis et al. (1998) found that some children (aged 9–10 years) demonstrated phonological priming while others did not. They suggested that individual differences may exist in the use of phonological information. Davis et al. did not speculate about the nature of those individual differences but, certainly, phonological skill per se can be measured and may be relevant.

Further, Plaut and Booth (2000) showed that differences in perceptual ability among college students predicted differences in semantic priming in a LDT. Greater semantic priming was found for those students with high perceptual ability. It seems possible that this individual difference might contribute to differences in early phonological priming.

Our speculation is that individual differences may predict whether or not a pseudohomophone priming effect is observed with a brief prime duration. Two individual difference measures, a measure of phonological

Table 3

Mean age, perceptual skill score, nonword naming reaction times (in ms) nonword naming error percentages and gender frequencies as a function of skill group

Perceptual skill group	Age	Male	Female	Perceptual skill score	
High ($n = 25$)	21.84	5	20	1.88	
Low ($n = 25$)	20.56	7	18	2.33	
Phonological skill group	Age	Male	Female	Naming RT	Naming error
High ($n = 25$)	21.24	8	17	656	11
Low ($n = 25$)	21.12	5	20	913	15

skill and a measure of perceptual skill, were included in this experiment.

3.1. Method

3.1.1. Participants

Fifty-one University of Calgary undergraduates participated in this experiment in exchange for bonus credit. All spoke English fluently. One participant was excluded from the analyses due to an overall error percentage of 40%. In order to divide participants into more and less skilled perceptual and phonological groups, each participant was rank ordered separately on their perceptual and phonological skill. A median split was performed on the ranked scores, which resulted in, for instance, 25 higher perceptual skill and 25 lower perceptual skill participants. The same process was followed for phonological skill. Mean skill scores and demographic characteristics for each skill group are presented in Table 3.

3.1.2. Materials

This experiment involved the same word list as Experiment 1b. As in Plaut and Booth (2000) perceptual skill was measured using the symbol search task from the Wechsler Intelligence Scale for Children (WISC-III). The symbol search task loads on the processing speed factor of the WISC-III. The task involves two sections, each with three pages of 15 items (90 items in total). In the first section participants must indicate as quickly and accurately as possible whether an arbitrary symbol on the left appears in a row of three arbitrary symbols on the right. In the second section participants are asked to indicate whether either of two symbols on the left appear in a row of five symbols on the right. On each trial, participants check either YES or NO boxes to the right of the symbols.

Phonological skill was assessed using a nonword naming task which included 43 nonwords chosen from the Word Attack subtest of the Woodcock Reading Mastery Tests Revised (1987). The nonwords (e.g., GAKED) ranged from one to two syllables and two to seven letters in length. Performance on this task was assessed by accuracy rate and RT for nonword pronunciation. Phonological skill groups were determined

by ranking each participant on accuracy rate and RT separately. Rankings were then summed to determine phonological skill.

3.1.3. Procedure

Participants first completed the symbol search task under a time limit of 120 s per section. The LDT was completed second. All procedures for the LDT were the same as in Experiment 1. Lastly, the participants completed the nonword naming task. Nonwords were presented using Psyscope on the same computer screen as the LDT. Participants were asked to pronounce each nonword as quickly and accurately as possible into a microphone that recorded naming latency. There was a 3 s intertrial interval between nonwords. Errors in pronunciation and mechanical errors were recorded by the experimenter.

3.2. Results

3.2.1. Overall analyses

Mean reaction times and error percentages for Experiment 2 are presented in Table 4. The mean error percentage for word targets was 14.41%. Once again, planned comparisons were conducted between priming conditions for mean reaction times and error percentages for both the monophthong and diphthong stimuli, and are presented in Table 4.

3.2.2. Perceptual skill

Mean reaction times and error percentages for participants high and low in perceptual skill are presented in Table 5. Means for monophthong stimuli are also illustrated in Fig. 1.

Monophthong. Our goal was to determine whether the advantage for the pseudohomophone prime condition over the orthographic prime condition was observed for more skilled participants. Hence, we conducted 2 (Prime Type: pseudohomophone, orthographic control) $\times$ 2 (Perceptual Skill: high, low) ANOVAs on the LDT data. Importantly, there was a significant interaction between Perceptual Skill and Prime Type in the RT analysis by subjects ($F_1(1, 48) = 5.21$, $p < .05$, $MSE = 2244.69$; $F_2(1, 47) = 3.15$, $p = .08$, $MSE = 4183.71$), but not in the error analysis ($F_1 < 1$; $F_2(1, 47) = 2.51$, $p > .05$,

Table 4

Mean reaction times (in ms), error percentages, and priming effects for overall analyses in Experiment 2 (all participants) as a function of prime type

Prime type	Target vowel complexity	RT		Error	
		Mean	SE	Mean	SE
Pseudohomophone	M	615	15.66	10.54	1.88
	D	603	17.01	14.40	1.46
Orthographic control	M	612	13.35	10.76	1.32
	D	601	11.69	7.96	0.96
Unrelated	M	657	14.82	23.40	2.19
	D	641	15.95	19.40	2.15
<i>Priming effects</i>					
Pseudohomophone vs. Orthographic	M	+3	Both $t < 1$	−0.22	Both $t < 1$
	D	+2	$t_1 < 1$ $t_2(47) = 1.29, p > .05$	+6.44	$t_1(49) = 3.94, p < .001$ $t_2(47) = 3.51, p < .001$
Orthographic vs. Unrelated	M	−45	$t_1(49) = -4.39, p < .001$ $t_2(47) = -3.29, p = .001$	−13.16	$t_1(49) = -6.18, p < .001$ $t_2(47) = -4.04, p < .001$
	D	−40	$t_1(49) = -3.83, p < .001$ $t_2(47) = -3.54, p < .001$	−11.44	$t_1(49) = -5.17, p < .001$ $t_2(47) = -3.95, p < .001$
Pseudohomophone vs. Unrelated	M	−42	$t_1(49) = -4.12, p < .001$ $t_2(47) = -2.97, p < .01$	−12.86	$t_1(49) = -6.36, p < .001$ $t_2(47) = -4.68, p < .001$
	D	−38	$t_1(49) = -2.82, p < .01$ $t_2(47) = -2.02, p < .05$	−5.00	$t_1(49) = -2.66, p < .01$ $t_2(47) = -1.92, p < .05$

Note. RT, reaction time; SE, standard error; M, monophthong; D, diphthong.

$MSE = .01$). These analyses also revealed a main effect of perceptual skill in the RT analysis ($F_1(1, 48) = 26.04, p < .001, MSE = 12398.13$; $F_2(1, 47) = 132.33, p < .001, MSE = 4935.66$), but not in the error analysis ($F_1 < 1$; $F_2(1, 47) = 1.19, p > .05, MSE = .01$) such that participants higher in perceptual skill responded faster than those lower in perceptual skill. There was no main effect of prime type in the RT analysis or in the error analysis ($F_s < 1$).

Planned comparisons were conducted between priming conditions for mean reaction times and error percentages for monophthong stimuli for participants high in perceptual skill and for participants low in perceptual skill, and these are presented in Table 5.

Diphthong. The 2 (Prime Type: pseudohomophone, orthographic control) $\times$ 2 (Perceptual Skill: high, low) ANOVAs revealed no significant interaction between Perceptual Skill and Prime Type in the RT analysis ($F_1(1, 48) = 1.71, p > .05, MSE = 3434.41$; $F_2 < 1$) or in the error analysis ($F_s < 1$). There was a main effect of perceptual skill in the RT analysis ($F_1(1, 48) = 21.62, p < .001, MSE = 12542.80$; $F_2(1, 47) = 138.23, p < .001, MSE = 3422.76$), but not in the error analysis ($F_1(1, 48) = 1.56, p > .05, MSE = .01$; $F_2(1, 47) = 2.01, p > .05, MSE = .01$) such that participants higher in perceptual skill responded faster than those lower in perceptual skill. There was no main effect of prime type in the RT analysis ($F_1 < 1$; $F_2(1, 47) = 2.15, p > .05, MSE = 4084.76$), however, there was an effect of prime type in the error analysis ($F_1(1, 48) = 15.22, p < .001, MSE = .01$; $F_2(1, 47) = 13.90, p < .01, MSE = .02$).

Planned comparisons were conducted between priming conditions for mean reaction times and error percentages for diphthong stimuli for participants high in perceptual skill and for participants low in perceptual skill, and these are presented in Table 5.

3.2.3. Phonological skill

Mean reaction times and error percentages for participants high and low in phonological skill are presented in Table 6. Means for monophthong stimuli are also illustrated in Fig. 2.

Monophthong. The 2 (Prime Type: pseudohomophone, orthographic control) $\times$ 2 (Phonological Skill: high, low) ANOVAs revealed a significant interaction between Phonological Skill and Prime Type in the RT analysis ($F_2(1, 47) = 17.16, p < .001, MSE = 5682.76$; $F_2(1, 47) = 5.15, p < .05, MSE = 5076.63$), and in the error analysis by items ($F_1(1, 48) = 3.10, p = .08, MSE = .01$; $F_2(1, 47) = 6.58, p < .05, MSE = .01$). There was no main effect of phonological skill in the RT analysis by subjects ($F_1(1, 48) = 2.51, p > .05, MSE = 18174.73$; $F_2(1, 47) = 17.16, p < .001, MSE = 5682.76$) or in the error analysis ($F_s < 1$). There was also no main effect of prime type in the RT analysis or in the error analysis ($F_s < 1$).

Planned comparisons were conducted between priming conditions for mean reaction times and error percentages for monophthong stimuli for participants high in phonological skill and for participants low in phonological skill, and these are presented in Table 6.

Table 5

Mean reaction times (in ms), error percentages, and priming effects for high and low perceptual skill participants in Experiment 2 as a function of prime type

Prime type	Target vowel complexity	RT		Error	
		Mean	SE	Mean	SE
<i>High perceptual skill</i>					
Pseudohomophone	M	547	13.61	8.28	1.30
	D	544	24.41	12.96	1.90
Orthographic control	M	566	15.11	10.20	1.87
	D	556	12.08	6.68	1.54
Unrelated	M	599	14.04	21.56	2.12
	D	582	15.76	14.84	2.00
<i>Low perceptual skill</i>					
Pseudohomophone	M	682	20.84	12.80	2.66
	D	663	16.99	15.84	2.48
Orthographic control	M	658	17.99	11.32	2.06
	D	645	15.75	9.24	1.96
Unrelated	M	715	20.41	25.24	3.01
	D	701	22.32	23.96	2.94
<i>High perceptual skill priming effects</i>					
Pseudohomophone vs. Orthographic	M	−19	$t_1(24) = -1.82, p < .05$ $t_2(47) = -1.03, p > .05$	−1.92	Both $t < 1$
	D	−12	Both $t < 1$	+6.28	$t_1(24) = 2.69, p < .01$ $t_2(47) = 2.76, p < .05$
Orthographic vs. Unrelated	M	−33	$t_1(24) = -2.58, p < .01$ $t_2(47) = -2.90, p < .01$	−11.36	$t_1(24) = -6.47, p < .001$ $t_2(47) = -3.19, p = .01$
	D	−26	$t_1(24) = -2.48, p < .05$ $t_2(47) = -2.69, p < .01$	−8.16	$t_1(24) = -3.65, p < .001$ $t_2(47) = -3.29, p = .001$
Pseudohomophone vs. Unrelated	M	−52	$t_1(24) = -5.13, p < .001$ $t_2(47) = -4.37, p < .001$	−13.28	$t_1(24) = -4.97, p < .001$ $t_2(47) = -4.30, p < .001$
	D	−38	$t_1(24) = -1.89, p < .05$ $t_2(47) = -1.57, p > .05$	−1.88	Both $t < 1$
<i>Low perceptual skill priming effects</i>					
Pseudohomophone vs. Orthographic	M	+24	$t_1(24) = 1.54, p > .05$ $t_2(47) = 1.14, p > .05$	+1.48	both $t < 1$
	D	+18	$t_1(24) = 1.47, p > .05$ $t_2(47) = 1.20, p > .05$	+6.60	$t_1(24) = 2.83, p < .01$ $t_2(47) = 3.07, p < .01$
Orthographic vs. Unrelated	M	−57	$t_1(24) = -1.83, p < .05$ $t_2(47) = -3.03, p < .01$	−13.92	$t_1(24) = -3.74, p < .001$ $t_2(47) = -3.85, p < .001$
	D	−56	$t_1(24) = -3.04, p < .01$ $t_2(47) = -2.93, p < .01$	−14.72	$t_1(24) = -3.92, p < .001$ $t_2(47) = -3.94, p < .001$
Pseudohomophone vs Unrelated	M	−33	$t_1(24) = -1.54, p > .05$ $t_2(47) = -1.41, p > .05$	−12.44	$t_1(24) = -4.03, p < .001$ $t_2(47) = -3.63, p < .001$
	D	−38	$t_1(24) = -2.09, p < .05$ $t_2(47) = -1.99, p < .05$	−8.12	$t_1(24) = -2.76, p < .01$ $t_2(47) = -2.44, p < .05$

Note. RT, reaction time; SE, standard error; M, monophthong; and D, diphthong.

Diphthong. The 2 (Prime Type: pseudohomophone, orthographic control) $\times$ 2 (Phonological Skill: high, low) ANOVAs showed no significant interaction between Phonological Skill and Prime Type in the RT analysis ($F_1(1, 48) = 1.32, p > .05, MSE = 3461.88; F_2 < 1$) or in the error analysis ($F_1(1, 48) = 3.25, p = .08, MSE = .01; F_2(1, 47) = 2.54, p > .05, MSE = .01$). There was also no main effect of perceptual skill in the RT analysis by subjects ($F_1(1, 48) = 2.62, p > .05, MSE = 17249.35; F_2(1, 47) = 21.90, p < .001, MSE = 3281.50$), or in the error analysis by subjects ($F_1(1, 48) = 1.71, p > .05,$

$MSE = .01; F_2(1, 47) = 4.18, p < .05, MSE = .01$). There was no main effect of prime type in the RT analysis ($F_1 < 1; F_2(1, 47) = 1.95, p > .05, MSE = 5086.58$), however, there was an effect of prime type in the error analysis ($F_1(1, 48) = 16.25, p < .001, MSE = .01; F_2(1, 47) = 11.59, p = .001, MSE = .02$).

Planned comparisons were conducted between priming conditions for mean reaction times and error percentages for diphthong stimuli for participants high in phonological skill and for participants low in phonological skill, and these are presented in Table 6.

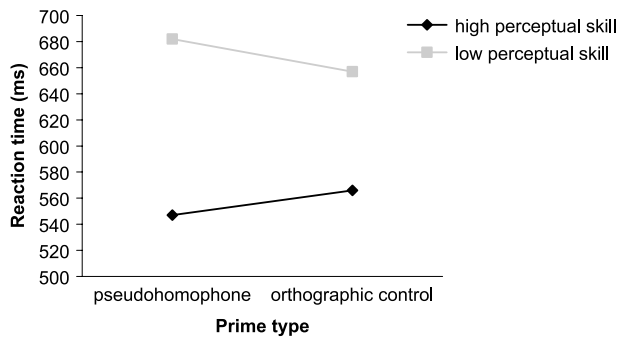


Fig. 1. Mean reaction times (in ms) for Experiment 2 as a function of prime type and perceptual skill.

4. General discussion

In the present Experiment 1, we were unable to find significant phonological priming in one LDT with 15-ms masked primes, but did observe significant phonological priming in a second LDT. The two experiments were virtually identical and yet one produced evidence for early phonological assembly while the other did not.

In an effort to reconcile these contrasting results, we included tests of perceptual and phonological skill in Experiment 2. These individual difference measures proved to be important, because in Experiment 2 for target words with a simple vowel structure, there were

Table 6

Mean reaction times (in ms), error percentages, and priming effects for high and low phonological skill participants in Experiment 2 as a function of prime type

Prime type	Target vowel complexity	RT		Error	
		Mean	SE	Mean	SE
<i>High phonological skill</i>					
Pseudohomophone	M	583	22.48	8.28	1.23
	D	575	28.57	11.52	2.14
Orthographic control	M	601	21.55	12.28	2.20
	D	586	18.39	7.96	1.59
Unrelated	M	629	17.95	21.88	2.67
	D	619	23.08	15.52	2.05
<i>Low phonological skill</i>					
Pseudohomophone	M	646	20.30	12.80	2.54
	D	631	17.32	17.28	2.30
Orthographic control	M	623	15.93	9.24	1.97
	D	615	14.22	7.96	1.72
Unrelated	M	685	22.52	24.92	3.17
	D	663	21.60	23.28	3.20
<i>High phonological skill priming effects</i>					
Pseudohomophone vs. Orthographic	M	-18	$t_1(24) = -1.45, p > .05$ $t_2(47) = -1.12, p > .05$	-4.00	$t_1(24) = -1.86, p < .05$ $t_2(47) = -1.39, p > .05$
	D	-11	$t_1 < 1$ $t_2(47) = -1.13, p > .05$	+3.56	$t_1(24) = 1.54, p > .05$ $t_2(47) = 1.65, p > .05$
Orthographic vs. Unrelated	M	-28	$t_1(24) = -2.21, p < .05$ $t_2(47) = -1.99, p < .05$	-9.60	$t_1(24) = -4.88, p < .001$ $t_2(47) = -2.80, p < .01$
	D	-33	$t_1(24) = -3.21, p < .01$ $t_2(47) = -2.89, p < .01$	-7.56	$t_1(24) = -3.53, p = .001$ $t_2(47) = -2.54, p < .01$
Pseudohomophone vs. Unrelated	M	-46	$t_1(24) = -3.80, p < .001$ $t_2(47) = -3.65, p < .001$	-13.60	$t_1(24) = -5.21, p < .001$ $t_2(47) = -4.47, p < .001$
	D	-44	$t_1(24) = -2.00, p < .05$ $t_2(47) = -2.15, p < .05$	-4.00	$t_1(24) = -1.70, p > .05$ $t_2(47) = -1.34, p > .05$
<i>Low phonological skill priming effects</i>					
Pseudohomophone vs. Orthographic	M	+23	$t_1(24) = 1.61, p > .05$ $t_2(47) = 1.89, p > .05$	+3.56	$t_1 < 1$ $t_2(47) = 1.29, p > .05$
	D	+16	$t_1(24) = 1.32, p > .05$ $t_2(47) = 1.33, p > .05$	+9.32	$t_1(24) = 4.21, p < .001$ $t_2(47) = 3.42, p < .001$
Orthographic vs. Unrelated	M	-62	$t_1(24) = -3.97, p < .001$ $t_2(47) = -3.56, p < .001$	-15.68	$t_1(24) = -4.45, p < .001$ $t_2(47) = -4.44, p < .001$
	D	-48	$t_1(24) = -2.58, p < .01$ $t_2(47) = -2.74, p < .01$	-15.32	$t_1(24) = -4.07, p < .001$ $t_2(47) = -3.60, p < .001$
Pseudohomophone vs. Unrelated	M	-39	$t_1(24) = -2.31, p > .05$ $t_2(47) = -1.80, p > .05$	-12.12	$t_1(24) = -3.86, p < .001$ $t_2(47) = -3.57, p < .001$
	D	-32	$t_1(24) = -2.01, p < .05$ $t_2 < 1$	-6.06	$t_1(24) = -2.01, p < .05$ $t_2(47) = -1.49, p > .05$

Note. RT, reaction time; SE, standard error; M, monophthong; and D, diphthong.

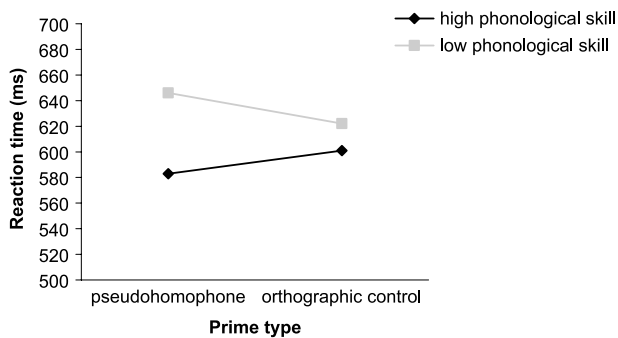


Fig. 2. Mean reaction times (in ms) for Experiment 2 as a function of prime type and phonological skill.

interactions between prime type and perceptual skill, and prime type and phonological skill. The nature of these interactions was that participants higher in perceptual and phonological skill tended to show a pseudohomophone priming effect, whereas participants lower in perceptual and phonological skill tended to be slower to respond to targets preceded by pseudohomophone primes in comparison to orthographic control primes. Therefore, individual differences in skill may provide an explanation for the previous discrepant findings in the literature regarding early phonological priming. It seems that it may be possible to reconcile the differential results of Lukatela and colleagues (Lukatela et al., 1998; Lukatela & Turvey, 2000), Forster and Guerreria-Mahoney (submitted), and Coltheart and Woollams (personal communication, January 22, 2003), among others (Ferrand & Grainger, 1992, 1993, 1994; Grainger & Ferrand, 1996), if one assumes that participants in those studies differed in perceptual and phonological skill.

We should note that, although both of the skill measures used in Experiment 2 were related to phonological priming effects, there is evidence that other individual difference measures are not related to phonological priming effects. In particular, in a previous study (Holyk & Pexman, 2002) we used reading skill as an individual difference measure and a procedure very similar to that used in the present study. We assessed reading skill with the Nelson–Denny reading test and the Author Recognition Test (Stanovich & West, 1989). Although those reading skill measures have been shown to be related to other visual word recognition effects (e.g., Chateau & Jared, 2000; Unsworth & Pexman, 2003) they were *not* related to early phonological priming effects.

The next question, then, is *why* perceptual skill and phonological skill are related to phonological priming effects. Our speculation is that those participants higher in perceptual skill are able to carry out more extensive processing on the prime and target under the difficult conditions of this experiment. Their ability to process visual information more quickly could allow them to make better use of the phonological information

provided by the prime. In contrast, there was little variability in participants' use of orthographic information in the primes (although, as noted in Footnote 1, our orthographic priming effects were not purely orthographic).

Participants higher in phonological skill also benefited more from the pseudohomophone primes than participants lower in phonological skill, but probably for slightly different reasons than those given for perceptual skill. Higher skill in phonological processing should lead to a greater efficiency and speed in assembling phonology (Frost, 1998). Therefore, participants higher in phonological skill will benefit more from the extra phonological information provided by the pseudohomophone prime than participants lower in phonological skill.

It is certainly possible that there is a common underlying variable that accounts for the similar findings regarding perceptual and phonological skill. If this were the case, however, one would expect that all participants who were high in phonological skill would also be high in perceptual skill. In actuality, 32 participants were either high or low on both measures, while 18 participants were low on one ranking and high on the other. Also, the correlation between perceptual skill scores and RT in the nonword naming task, while positive, was not significant ($r(49) = .26, p > .05$). In future research, the variables underlying the present individual difference effects should be explored.

In the present study, a pseudohomophone priming effect was only observed for monophthong (simple vowel) target words and never for diphthong (complex vowel) target words. This is consistent with Lukatela and Turvey's (2000) results. Lukatela and Turvey argued that phonological priming at such a brief prime duration would only be observed for words with a simple phonological translation. Although this could be the case, it is also the case that all of the monophthong target words began with the letter C and their pseudohomophone primes all started with the letter K (e.g., KREST–*crest*). This was not the case for diphthong targets. This peculiarity in the stimuli creates the possibility that significant phonological priming with a 15-ms prime exposure may depend on the C–K relationship. Future research should involve similar investigations using lists different from the C–K pseudohomophone prime–target pairings. If such an early phonological priming effect also appears with other stimulus lists it would strengthen the argument that phonological activation occurs at the earliest stages of word recognition.

Relatedly, while the pseudohomophone primes for monophthong targets always replace the initial consonant (using the C–K pairing), the pseudohomophone primes for diphthong targets replaced letters in the word body and left the initial letter unchanged. These differences between the pseudohomophone primes for the

monophthong and diphthong stimuli are important because Lukatela and Turvey (2000) only found significant pseudohomophone priming for monophthong stimuli. These differences could be responsible for the differential priming observed and cast some doubt on the conclusion that the added phonological complexity of the vowel for diphthongs is the reason no phonological priming was observed for those words.

It should be noted that the interpretation of results using very short prime durations is somewhat problematic. Frost and Yogeve (2001) warned researchers against using exposure duration as a reflection of the exact time course of phonological versus orthographic computation. There are many factors other than the set exposure duration that affect the perceptual experience of a participant in a fast priming experiment, all of which have the capability of shifting the results to completely opposite theoretical interpretations on the part of the researcher. The factors identified by Frost and Yogeve include the type of mask, whether the mask is a forward or backward mask, brightness of the screen, overall luminance, ambient luminance, font size of the words, font color and background color, distance of the participant from the monitor, and the refresh rate of the computer monitor. Given the above considerations and the fact that the usual prime durations of experiments examining early phonological and orthographic

computation range from 14 to 60 ms, Frost and Yogeve argued that extreme caution should be taken in the interpretation and comparison of results.

Frost and Yogeve (2001) have argued that, nonetheless, experiments using the fast priming paradigm can still provide good approximations of very early processing. That is, even though it is difficult to speak of absolute time courses, these experiments can provide evidence for strong claims about relative processing. If there is evidence of orthographic priming and no evidence of phonological priming using a 15 ms prime, while there is evidence of phonological priming using a 30 ms prime, the researcher can fairly confidently claim that orthographic information is processed first relative to phonological information.

5. Conclusion

The evidence presented in this paper does not provide irrefutable evidence for or against early phonological priming. The results of the present study do suggest, however, that in addition to the list of factors identified by Frost and Yogeve (2001), phonological priming effects are also sensitive to individual differences in perceptual skill and phonological processing.

Appendix A

Monophthong stimuli

Target	Pseudohomophone	Orthographic control	Unrelated
clip	KLIP	CLEP	FERD
cord	KORD	CYRD	LIRT
cream	KREAM	CROAM	BOURT
crisp	KRISP	CRASP	EIVEN
clerk	KLERK	CLORK	VABES
clock	KLOCK	CLECK	BRIND
creek	KREEK	CROEK	MOULP
catch	KATCH	COTCH	BROIK
card	KARD	CORD	BLUM
crust	KRUST	CROST	SMECK
clue	KLUE	CLOE	PRAN
cling	KLING	CLONG	PODES
clam	KLAM	CLOM	ERST
cans	KANS	CYNS	DARM
cash	KASH	COSH	WERP
class	KLASS	CLOSS	WHIKE
clap	KLAP	CLOP	HORT
cross	KROSS	CRUSS	BLAIF
crush	KRUSH	CRESH	SPORK
cubs	KUBS	CIBS	NEND
clan	KLAN	CLIN	SORP
cars	KARS	CORS	DINT
craft	KRAFT	CRIFT	NENCH

Appendix A (*continued*)

Target	Pseudohomophone	Orthographic control	Unrelated
cram	KRAM	CRIM	SNOX
clot	KLOT	CLAT	TASP
cramp	KRAMP	CROMP	SNOIT
crab	KRAB	CROB	POSK
cart	KART	CORT	BESK
crack	KRACK	CRYCK	TWINT
clump	KLUMP	CLIMP	STOIR
crash	KRASH	CRISH	PLEEM
cliff	KLIFF	CLOFF	SARTY
cork	KORK	CARK	NEAP
cult	KULT	CALT	LESP
creed	KREED	CROED	POASK
crop	KROP	CREP	FEAN
click	KLICK	CLECK	DOART
camp	KAMP	CUMP	TORD
corn	KORN	CURN	FLID
clog	KLOG	CLEG	SARK
cats	KATS	CYTS	NOIR
club	KLUB	CLEB	SARM
crap	KRAP	CRIP	BIST
cuts	KUTS	CITS	GARL
cups	KUPS	CIPS	FALM
crib	KRIB	CROB	VOLK
cast	KASt	CUST	DERG
crest	KREST	CROST	GOAMP

Diphthong stimuli

Target	Pseudohomophone	Experiment 1a orthographic control	Experiments 1b and 2 orthographic control	Unrelated
name	NAIM	NALM	NAML	PLIP
rope	ROAP	ROSP	ROPS	MANT
bake	BAIK	BAWK	BAKW	NERT
rain	RANE	RANS	RASN	KEED
wait	WATE	WATH	WAHT	RESP
tone	TOAN	TOON	TOND	PAIM
joke	JOAK	JONK	JOKN	MARN
fail	FALE	FALS	FASL	SECK
oats	OTES	OTRS	ORTS	GERP
fate	FAIT	FANT	FATN	CRIM
roar	RORE	RORT	ROTR	HEST
fake	FAIK	FAK	FAKS	NILP
blade	BLAID	BLARD	BLADR	GROIP
stove	STOAV	STOOV	STOVT	THARN
safe	SAIF	SARF	SAFR	LIRM
frame	FRAIM	FRALM	FRAML	BLORK
game	GAIM	GALM	GAML	RILP
boat	BOTE	BOTS	BOST	TINE
soak	SOKE	SOKL	SOLK	REND
hate	HAIT	HANT	HATN	BOIK
role	ROAL	ROTL	ROLT	NISM
goal	GOLE	GOLK	GOKL	DREN
stone	STOAN	STORN	STONR	KAIRN

Appendix A (continued)

Target	Pseudohomophone	Experiment 1a orthographic control	Experiments 1b and 2 orthographic control	Unrelated
trade	TRAID	TRALD	TRADL	SMILN
lake	LAIK	LASK	LAKS	HURB
hole	HOAL	HORL	HOLR	LESK
tape	TAIP	TASP	TAPT	FLIN
grade	GRAID	GRALD	GRADL	STOIM
rate	RAIT	RAST	RATP	TING
hope	HOAP	HOSP	HOPK	CHIB
soap	SOPE	SOPT	SOTP	VELM
mate	MAIT	MANT	MATN	GRIB
home	HOAM	HORM	HOMR	PARD
late	LAIT	LANT	LATN	PRUD
bone	BOAN	BOON	BONT	LARM
pole	POAL	PORL	POLR	SWIR
rake	RAIK	RWAK	RWKE	WHOB
foam	FOME	FOMP	FOPM	SCUP
goat	GOTE	GOTS	GOST	TWEN
rail	RALE	RALD	RADL	FESK
fame	FAIM	FALM	FAML	TOLG
mode	MOAD	MOND	MODN	NARP
gain	GANE	GAND	GADN	SOTH
date	DAIT	DAST	DATS	FISK
vote	VOAT	VONT	VOTN	MARG
same	SAIM	SARM	SAMR	RINP
train	TRANE	TRANT	TRATN	GOADE
take	TAIK	TARK	TAKR	GRUD

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